

Zero-Shot Style Extraction and LoRA Distillation from a Pretrained Text-to-Image Model

A Feasibility Study on Extracting Model-Internal Style Representations from Krea 2 RAW

Abstract

LoRA (Low-Rank Adaptation) has become one of the most widely used methods for adapting text-to-image diffusion models to a specific visual style. Conventional style LoRA training, however, generally requires a curated dataset of images paired with captions or prompts. This introduces a significant dependency on image collection, preprocessing, captioning, and GPU-based training.

This paper proposes a different approach: zero-shot style extraction from the internal knowledge of an already pretrained text-to-image model.

The central hypothesis is that a sufficiently capable text-to-image foundation model already contains latent representations of many visual styles. Therefore, rather than teaching a model a style from external images, it may be possible to identify the model's existing response to a style concept and convert that response into a reusable parameter-efficient adapter.

We propose a framework based on Krea 2 RAW. For a given content prompt P_c and a corresponding content-plus-style prompt $P_{\{c+s\}}$, the model is evaluated under identical latent states and timesteps. Because Krea 2 is a flow-matching model, its prediction can be represented as a velocity field:

$$[v_c = F_{\theta}(x_t, t, P_c)]$$

$$[v_{c+s} = F_{\theta}(x_t, t, P_{c+s})]$$

The difference

$$[D_s = v_{c+s} - v_c]$$

is treated as a candidate style-induced behavioral direction.

By collecting these differences across multiple semantic concepts, random seeds, and timesteps, we investigate whether a stable, content-independent style subspace can be identified. If such a direction exists, we propose to distill it into a low-rank parameter update:

$$[\Delta W \approx BA]$$

where BA represents a LoRA update.

The resulting adapter could then be applied to arbitrary prompts without explicitly mentioning the original style concept. The final visual effect can be evaluated externally using an image-generation platform such as Tensor.Art.

The proposed method does not claim that arbitrary new styles can be created without data.

Instead, it investigates whether existing model knowledge can be reorganized into a new default behavior without conventional image-based fine-tuning.

1. Introduction

1.1 Background

Modern text-to-image foundation models acquire a broad range of visual concepts during pretraining. These concepts include objects, environments, artistic techniques, photographic characteristics, rendering styles, composition patterns, and associations between language and visual appearance.

A conventional approach to creating a style

LoRA is to collect a set of images representing the desired style and optimize a small set of additional parameters so that the model reproduces that style.

This process can be summarized as:

[\text{Images} \rightarrow \text{Training} \rightarrow \text{LoRA}]

Although effective, this pipeline has several limitations:

1. A suitable image dataset is required.
2. Image-caption quality affects training.
3. Dataset curation is time-consuming.
4. Training generally requires substantial GPU resources.
5. The resulting adapter may partially memorize characteristics of the training images.
6. It is difficult to determine whether the adapter represents a general style concept or merely the statistical properties of the

dataset.

This work investigates an alternative:

[\text{Model knowledge} \rightarrow \text{Style extraction} \rightarrow \text{LoRA}]

The key question is whether a pretrained model can serve as its own source of supervision.

2. Research Question

The primary research question is:

Can an existing text-to-image foundation model generate a reusable LoRA style adapter by comparing its own responses to content prompts and content-plus-style prompts, without using external image training data?

This can be divided into several subquestions.

RQ1

Does a pretrained model exhibit a consistent internal response to a style concept across different semantic contents?

RQ2

Can the difference between a content-only prediction and a content-plus-style prediction be isolated as a meaningful style direction?

RQ3

Can this style direction be approximated by a low-rank parameter update?

RQ4

Can the resulting LoRA transfer the extracted style to prompts that do not explicitly contain the style word?

RQ5

Can this process be performed as a CPU-oriented research pipeline by avoiding image decoding and minimizing repeated model computation?

3. Hypothesis

We propose the following hypothesis:

H1: If a pretrained text-to-image model already contains a sufficiently well-defined representation of a visual style, the model's response difference between a content prompt and a content-plus-style prompt contains a reusable style-dependent component that can be approximated by a

low-rank parameter update.

Formally, let:

[$P_c = \text{content prompt}$]

and

[$P_{c+s} = \text{content + style prompt}$]

Then:

[$D_s = F_{\theta}(x_t, t, P_{c+s})$

$F_{\theta}(x_t, t, P_c)$]

is expected to contain both style-related and residual content-dependent components:

[$D_s = D_{\text{style}} + D_{\text{residual}}$]

The objective of the proposed method is therefore not simply to calculate one difference, but to estimate a style component that remains stable across many contents:

[$\hat{D}_s = \mathbb{E}_{c,z,t} [F_{\theta}(x_t, t, P_{c+s})$

$F_{\theta}(x_t, t, P_c)$]

where c represents semantic content, z represents the random seed or latent state, and t represents the flow-matching timestep.

4. Why Krea 2 RAW?

This research specifically considers Krea 2 RAW rather than a conventional diffusion checkpoint. Krea 2 is a modern text-to-image foundation model based on a dense 12B-parameter DiT architecture. Its RAW variant is particularly relevant because it is designed to retain a high degree of diversity and malleability and is intended for downstream adaptation such as fine-tuning and LoRA training.

Krea 2 uses a flow-matching formulation rather

than the conventional epsilon-prediction formulation used by many earlier diffusion models.

Therefore, the proposed method operates on the model's predicted velocity:

$$[v_{\theta}(x_t, t, c)]$$

rather than treating the output as a conventional noise prediction.

This makes Krea 2 RAW an interesting testbed for investigating whether model-internal semantic and stylistic behavior can be directly measured in the model's vector field.

5. Proposed Method

5.1 Overview

The proposed framework consists of five stages:

1. Style prompting
2. Response differencing
3. Style subspace extraction
4. Low-rank parameter distillation
5. External image validation

The complete pipeline is:

[\boxed{ \text{Prompt Pairing} \rightarrow \text{Velocity Difference} \rightarrow \text{Style Subspace} \rightarrow \text{LoRA Distillation} \rightarrow \text{Image Validation} }]

6. Stage 1 — Prompt Pair Construction

For a target style 錒s錒, each experiment uses

two prompts.

Content-only prompt

[P_c]

Content-plus-style prompt

[P_{c+s}]

For example:

text



a tiger walking through a jungle

versus:

text



a watercolor tiger walking through a
jungle

Additional concepts may include:

text



a woman standing in a city
a spaceship above a planet
a medieval castle on a mountain
a bowl of fruit on a wooden table
a futuristic motorcycle in a desert

Each concept is paired with the same style expression.

To reduce sensitivity to a particular wording, multiple style formulations can eventually be used:

text



watercolor
watercolor painting
traditional watercolor illustration
hand-painted watercolor

This allows the experiment to distinguish the

underlying style concept from a single textual token.

7. Stage 2 — Matched Model Evaluation

The most important experimental control is that the two prompts should be evaluated using the same underlying state.

For a given latent x_t and timestep t :

$$[v_c = F_{\theta}(x_t, t, P_c)]$$

and:

$$[v_{c+s} = F_{\theta}(x_t, t, P_{c+s})]$$

The candidate style response is:

$$[D_s = v_{c+s} - v_c]$$

The use of the same latent state and timestep is important because otherwise the observed

difference could contain random variation unrelated to the style prompt.

The proposed experiment therefore treats the paired prediction as a controlled intervention:

[\text{Style prompt} \rightarrow \text{change in model behavior}]

8. Stage 3 — Extracting a Stable Style Direction

A single prompt pair is insufficient to establish that D_s represents style.

For this reason, many different semantic concepts are evaluated.

For each concept c_i :

$$[D_i = F_{\theta}(x_t, t, P_{\{c_i+s\}})$$

$F_{\theta}(x_t, t, P_{\{c_i\}})$]

The resulting vectors are analyzed for consistency.

8.1 Cosine Similarity

A simple first test is pairwise cosine similarity:

$$\cos(D_i, D_j)$$

$$\frac{D_i \cdot D_j}{\|D_i\| \|D_j\|}$$

If the style effect is genuinely shared across concepts, we expect higher similarity between style-induced differences than would be expected from unrelated model perturbations.

8.2 Mean Style Direction

A first approximation is:

$$[\bar{D}_s = \frac{1}{N} \sum_{i=1}^N D_i]$$

This produces a global style direction.

However, simple averaging may be insufficient because different concepts may contain different residual effects.

8.3 PCA / Style Subspace

A more general approach is to perform PCA on the collection:

$$[D_1, D_2, \dots, D_N]$$

If the first principal component explains a substantial portion of the variance, this may indicate the existence of a low-dimensional style subspace.

Instead of representing style using a single vector, we can therefore define:

$$[\mathcal{S}_s = \operatorname{span}\{$$

$(u_1, \dots, u_k)$

where u_i are the principal components.

This may be more robust than a single averaged direction.

9. Stage 4 — From Style Direction to LoRA

Finding a style direction is only the first half of the problem.

The next question is whether this behavior can be encoded as a parameter-efficient weight update.

Let the original model parameters be:

θ

and the desired updated parameters be:

$\theta' = \theta + \Delta\theta$

The ideal update should satisfy:

$$[F_{\theta + \Delta \theta}(x_t, t, P)$$

$$F_{\theta}(x_t, t, P) \approx D_s]$$

for a broad range of prompts P .

Instead of learning an unrestricted

$\Delta \theta$, we constrain the update to a low-rank form.

For a weight matrix W :

$$[W' = W + BA]$$

where:

$$[B \in \mathbb{R}^{d \times r}]$$

and:

$$[A \in \mathbb{R}^{r \times k}]$$

with:

$$[r \leq \min(d, k)]$$

This is the standard LoRA parameterization.

10. Style-Direction Distillation Objective

The proposed LoRA can be optimized using the model itself as the teacher.

For a prompt P , define:

$$\Delta v_{\text{LoRA}}$$

$$F_{\theta+BA}(x_t, t, P)$$

$$F_{\theta}(x_t, t, P)$$

The objective is:

$$\mathcal{L}_{\text{style}}$$

$$\left| \Delta v_{\text{LoRA}} \right|$$

$$\alpha D_s \left| \right|^2$$

where α controls the desired style strength.

The LoRA therefore learns:

"When this adapter is active, modify the model's behavior in approximately the same direction as adding the target style concept to the prompt."

This is fundamentally different from conventional LoRA training.

Conventional training:

[\text{image} \rightarrow \text{target behavior}]

Proposed training:

[\text{model response difference} \rightarrow \text{target behavior}]

11. Direct Weight Editing as an

Alternative

An alternative formulation is to first estimate an unrestricted parameter update:

$$[\Delta W]$$

and then approximate it using low-rank decomposition.

For example:

$$[\Delta W \approx U \Sigma V^T]$$

Taking the first r singular components:

$$[\Delta W_r]$$

$$U_r \Sigma_r V_r^T]$$

we can construct:

$$[B = U_r \Sigma_r]$$

$$[A = V_r^T]$$

resulting in:

$$[\Delta W_r = BA]$$

This creates a conceptual bridge between:
model editing

and

LoRA distillation.

The unrestricted model edit may reveal whether the desired behavioral change is inherently low-rank before attempting to constrain it to a small LoRA.

12. Which Layers Should Be Modified?

One major experimental question is where the style information is represented.

Rather than assuming a particular layer in advance, several target configurations can be compared.

Experiment A

Attention layers only.

Experiment B

MLP layers only.

Experiment C

Text-fusion layers only.

Experiment D

Attention + text-fusion.

Experiment E

Attention + MLP + text-fusion.

The purpose is to determine whether style behavior is localized or distributed throughout the network.

A particularly interesting hypothesis is that style-conditioned changes may be strongly associated with the interaction between textual

conditioning and the transformer representation. However, this should be treated as an experimental hypothesis rather than an established property of Krea 2.

13. CPU-Only Experimental Design

An important practical constraint of this research is that the local experimental environment may not contain a GPU.

Therefore, the proposed pipeline intentionally separates model inference from image generation.

The local machine does not need to generate final images.

Instead:

text



Krea 2 RAW



forward pass



velocity tensors



save to disk



PCA / statistics / fitting



LoRA

Image decoding is unnecessary during the initial experiments.

This substantially reduces the computational requirements of the research pipeline compared with conventional image-generation experiments. The most expensive operation remains the Krea 2 RAW forward pass. Therefore, the experimental design should minimize the

number of model evaluations.

A small proof-of-concept could begin with:

- 10–20 semantic concepts
- 2–5 random seeds
- 4–8 timesteps
- one target style
- no image decoding

The resulting tensors can be cached and reused for subsequent analysis.

14. External Image Validation

Because the local machine is CPU-only, the final visual evaluation can be performed using an external inference platform such as Tensor.Art. The evaluation should contain at least four

conditions.

Condition A — Baseline

Prompt:

text



a girl walking through a forest

No LoRA.

Condition B — Explicit Style

Prompt:

text



a watercolor girl walking through a forest

No LoRA.

This establishes that the base model already responds to the style concept.

Condition C — Proposed LoRA

Prompt:

text



a girl walking through a forest

With the proposed LoRA.

The style word is intentionally removed.

Condition D — Combined

Prompt:

text



a watercolor girl walking through a forest

With the proposed LoRA.

These four conditions allow several questions to be separated.

Most importantly:

[A $\rightarrow$ C]

tests whether the LoRA creates a default style behavior.

Meanwhile:

[B $\rightarrow$ C]

tests whether the LoRA reproduces a behavior already accessible through the model's native style concept.

15. Generalization Test

A successful result should not depend on the concepts used during extraction.

Therefore, training concepts and evaluation concepts should be separated.

For example:

Extraction concepts

text



tiger

castle

woman

spaceship

fruit

motorcycle

Evaluation concepts

text



dog

architecture

portrait

train

landscape

robot

food

The LoRA should then be evaluated without

explicitly mentioning the target style.

If the style appears consistently across unseen semantic categories, this provides stronger evidence that the adapter represents a general style behavior rather than concept-specific memorization.

16. Measuring Style Consistency

Visual inspection alone may not be sufficient. Several quantitative measurements can be explored.

16.1 Directional Consistency

Measure:

$$[\operatorname{cos}(D_i, D_j)]$$

across different concepts.

16.2 Rank Concentration

Perform PCA/SVD and calculate the explained variance ratio of the leading components.

If:

$$\left[\frac{\lambda_1}{\sum_i \lambda_i} \right]$$

is large, the style-induced behavior may be approximately low-dimensional.

16.3 Prompt Independence

Evaluate the generated images across unrelated semantic categories.

A good style adapter should maintain stylistic

consistency while allowing content to change.

16.4 Style Strength

Generate images at different LoRA weights:

[0.25,\ 0.5,\ 0.75,\ 1.0,\ 1.25]

and determine whether the visual style changes monotonically with adapter strength.

17. Expected Advantages

If successful, the proposed method could provide several advantages.

17.1 No Image Dataset

The method does not require collecting a dedicated style dataset.

17.2 Reduced Annotation Requirements

No image-caption pairs are required for the initial style extraction stage.

17.3 Model-Internal Supervision

The pretrained model acts as its own teacher.

17.4 Potentially Lower Training Cost

The method may require substantially fewer training examples than conventional image-based LoRA training.

17.5 Better Understanding of Model Representations

The method can be used to investigate where and how visual styles are represented inside a

text-to-image foundation model.

17.6 Portable Output

The final result is a LoRA adapter rather than a complete modified checkpoint.

18. Limitations

The proposed method has several important limitations.

18.1 It Cannot Create Completely Unknown Styles

The method depends on the pretrained model already possessing some representation of the target style.

If the model has no meaningful knowledge of a

style:

[\text{No internal representation} \rightarrow \text{No reliable style direction}]

The method therefore reorganizes existing knowledge rather than creating entirely new knowledge from nothing.

18.2 Prompt Semantics May Contaminate the Difference

The difference:

[$v_{c+s} - v_c$]

may contain more than visual style.

The style token can alter:

- composition
- object appearance
- semantics

- lighting
- texture
- color
- spatial relationships

Therefore, cross-concept averaging and subspace analysis are essential.

18.3 The Style Direction May Not Be Globally Linear

A single vector may not adequately represent style across all semantic contexts.

The actual behavior could be:

$$[D_s = f(c, s, t)]$$

rather than:

$$[D_s = f(s)]$$

This motivates the use of style subspaces and

potentially nonlinear parameterization.

18.4 LoRA Rank May Be Insufficient

Even if the desired behavior exists in the model, it may not be representable using a very small rank.

The experiment should therefore compare ranks such as:

$[r \in \{4, 8, 16, 32\}]$

rather than assuming rank 4 or 8 is sufficient.

18.5 RAW-to-Turbo Transfer Requires Validation

Krea 2 provides RAW and Turbo variants with different inference characteristics.

Although Krea's intended workflow supports training LoRAs on RAW and applying them to Turbo, a LoRA produced through this proposed zero-shot distillation method should still be empirically tested for cross-checkpoint compatibility.

19. Proposed Experimental Roadmap

The research can be divided into four phases.

Phase 1 — Style Direction Discovery

Goal:

Determine whether a stable style direction

exists.

Tasks:

1. Select one known style.
2. Generate 10–20 prompt pairs.
3. Use matched latent/timestep conditions.
4. Extract velocity differences.
5. Calculate cosine similarity.
6. Perform PCA.

Success criterion:

A meaningful cross-concept common direction or low-dimensional subspace is observed.

Phase 2 — Layer Localization

Goal:

Determine where the style-induced behavior is

represented.

Tasks:

- capture internal activations
- compare prompt-pair differences
- evaluate attention
- evaluate MLP
- evaluate text-fusion
- compare transformer blocks

Success criterion:

Identify one or more regions where style-related variation is significantly more consistent than elsewhere.

Phase 3 — LoRA Distillation

Goal:

Convert the extracted style behavior into a portable adapter.

Tasks:

1. Freeze Krea 2 RAW.
2. Initialize LoRA.
3. Optimize against the extracted style direction.
4. Compare different ranks.
5. Compare different target modules.
6. Save .safetensors.

Success criterion:

The LoRA reproduces a significant portion of the desired model-response change.

Phase 4 — External Generation

Goal:

Determine whether the numerical behavior produces an actual visual style.

The LoRA is uploaded to Tensor.Art.

Test:

text



content only

+

LoRA

without explicitly mentioning the style.

Success criterion:

The target visual style consistently appears across previously unseen prompts.

20. Baselines

To establish whether the proposed method provides a meaningful advantage, several baselines should be considered.

Baseline 1 — No LoRA

Original Krea 2.

Baseline 2 — Explicit Style Prompt

Original Krea 2 + style word.

Baseline 3 — Conventional Style LoRA

If a suitable image dataset is available, train a conventional LoRA and compare its behavior.

Baseline 4 — Random / Unrelated LoRA

Tests whether observed effects are simply caused by introducing any adapter.

Baseline 5 — Simple Weight-Space SVD

Compare direct parameter modification with the proposed behavioral distillation approach.

21. Expected Contributions

This work proposes three potential contributions.

Contribution 1

A framework for extracting a visual style representation from a pretrained text-to-image model without requiring external style images.

Contribution 2

A method for converting model-internal style behavior into a low-rank LoRA adapter.

Contribution 3

An experimental framework for studying the relationship between language-conditioned model behavior, style representations, and parameter-efficient adaptation.

The work may therefore be relevant not only to LoRA training, but also to broader questions concerning:

- model editing
- concept manipulation
- representation engineering
- parameter-efficient fine-tuning
- latent style representations
- text-to-image foundation model interpretability

22. Discussion

The central idea of this research is to change the role of the pretrained model.

In conventional fine-tuning, the model is the student and an external image dataset is the teacher:

[\text{Dataset} \rightarrow \text{Model}]

In the proposed framework, the model itself becomes both the source of knowledge and the teacher:

[\text{Pretrained Model} \rightarrow
 \text{Behavior Extraction} \rightarrow
 \text{LoRA Student}]

This creates a form of self-distillation at the parameter level.

The model already knows how to respond when the user writes:

"watercolor"

The research question is whether we can observe the difference between:

"a tiger"

and:

"a watercolor tiger"

and then encode that difference into the model's parameters so that:

"a tiger"

behaves approximately like:

"a watercolor tiger"

without requiring the user to explicitly provide the style concept.

This can be interpreted as converting a conditional behavior into an adapter-controlled prior.

23. Broader Interpretation

If the experiment succeeds, the result would suggest that certain textual concepts in a text-to-image model can be transformed into parameter-space behavior.

For example:

["watercolor" $\rightarrow$
 $D_{\{\text{watercolor}\}}$ $\rightarrow \Delta W$
 $\rightarrow \text{LoRA}$]

Similarly, the same framework could potentially be tested on:

- cinematic photography
- oil painting
- anime-like rendering
- pencil sketch
- vintage photography
- concept art
- editorial illustration

- 3D rendering

The important distinction is that the proposed method does not attempt to train these styles from scratch.

Instead, it asks:

Can existing model knowledge be extracted, isolated, and re-encoded into a new parameter-efficient behavior?

24. Conclusion

This paper proposes a zero-shot framework for extracting visual style behavior from a pretrained text-to-image foundation model and distilling that behavior into a LoRA adapter without using external image training data.

Using Krea 2 RAW as the experimental model,

the proposed method compares model responses to matched content-only and content-plus-style prompts. Because Krea 2 uses flow matching, the difference between the corresponding velocity predictions provides a natural candidate representation of the model's style-induced behavioral change.

By collecting these differences across semantic concepts, seeds, and timesteps, the method attempts to identify a stable style direction or low-dimensional style subspace. This representation can then serve as a behavioral target for low-rank parameter optimization.

The resulting LoRA can be evaluated externally through an image-generation platform such as Tensor.Art, with particular emphasis on whether the target style appears when the style word is removed from the prompt.

The proposed research therefore explores a potentially useful alternative to conventional

dataset-driven LoRA training:

$$[\boxed{\text{Existing Model Knowledge}} \rightarrow \text{Style Direction} \rightarrow \text{Behavior Distillation} \rightarrow \text{LoRA}]$$

The primary feasibility question is not whether the method can generate an aesthetically pleasing image, but whether a pretrained model's own internal style response can be isolated sufficiently well to become a reusable, content-independent, low-rank parameter modification.

If demonstrated, this would provide evidence that some forms of visual style adaptation can be achieved by reorganizing knowledge already present in a foundation model rather than learning the style from an external image dataset.